G H Raisoni International Skill Tech University, Pune.

Question Paper

Continuous Assessment Examination-II(CAE-II)

Year: First **Semester: II**

Programme: B. Tech **Duration: 60 mins.**

Max Marks: 20 **Course Code: 24BTET205**

Course Name: Programming Logic Skills

Instructions:

1. Follow the question numbering.
2. Read all instructions carefully.
3. Students must label their answers with the corresponding question number, such as Q.No. 1A, Q.No. 2C, etc.

Note: This question paper addresses below Course Outcomes (CO's):

CO3	Demonstrate understanding of sets, functions, sequences, matrices, and use it in computational scenarios
CO4	Compare various types of programming languages and understand the process of program development.

Q. No.	Sub Q. No.	Question	Marks
1		**Solve any TWO**	
	A	Define Set. Describe Union and Intersection with venn diagram.	1+2+2=5
	B	Find the inverse of following functions 1) $f(x) = \frac{1+9x}{4-x}$ 2) $f(x) = \frac{6-10x}{8x+7}$	5
	C	Find 1) fog(x) 2) gog(x) where f(x)=2x+3 , g(x)=3x+2	2+3=5
	D	Find A+B and AxB for given matrices. $A = \begin{bmatrix} 4 & 0 & 2 \\ 1 & 2 & 1 \\ 2 & 4 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 3 & 1 & 0 \end{bmatrix}$	2+3=5
2		**Solve any TWO**	
	A	Differentiate between compiler and interpreter.	5
	B	Compare procedural and object-oriented programming language with examples.	5
	C	What is the difference between syntax and semantics in programming languages?	5
	D	Explain the phases of the PDLC.	5

--- End of the Question Paper ---